TER - Proposition de Sujet
Cost-Sensitive Learning for Imbalanced Problems

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Context: With the widespread use of Machine Learning in various real settings, several problems that were not necessarily prominent in controlled environments become more important. One of them is imbalanced learning in binary classification where one of the classes, usually the positive one, is far less represented in the dataset than the other. For example, this problem arises in fraud or anomaly detection in banking: in this case frauds or anomalies are extremely minor and represent less than 1% of the data. It is also encountered in other contexts such as economic, health, security and is at the heart of research topics of large organizations, for example the detection of tax frauds by the French Ministry of Economy.

This is a complex problem that may be very challenging for standard classification algorithms. Indeed, the aim of the latter is usually to minimize the error rate of the learned model and, in an imbalanced setting, this can be trivially achieved by a constant classifier that always predicts the majority class. In the aforementioned fraud detection example, such a model would correctly predict 99% of the examples but will not achieve the original goal: detecting frauds. This shows that it is necessary to design algorithms able to work with other performance measures that are more adapted to this context.

An example of such an alternative measure is the F-measure. It is able to take into account both the ability of the model to find examples of the minority class and the confidence of the model in its predictions of belonging to this same class. It is defined as:

$$F = \frac{2(P - FN)}{2P - FN + FP} = \frac{2(P - e_1)}{2P - e_1 + e_2},$$

where P is the number (or proportion) of positives in the dataset, $e_1 = FN$ and $e_2 = FP$ are respectively the number (or proportion) of *False Negatives* and *False Positives* predicted by the model. Unfortunately, maximizing such a measure is a difficult task due to non convexity of the underlying problem. Nevertheless, several solutions have been proposed in the literature, for example using its *fractional linear* property[1] (Parambath et al., 2014).

Objective: Another approach to optimize the F-measure was proposed by Bascol et al. (2019). This method uses theoretical upper bounds on the optimal F-measure of a model and its neighbours to design an iterative algorithm that progressively improves the performances of the learned model. This is illustrated in Figure 1. One of the limits of this approach is that it only considers theoretical upper bounds but no lower bounds on the F-measure. It means that it looks for models that could potentially perform very well (with a high upper bound) but does not take into account the fact that they could perform very poorly (if they have a low lower bound), this illustrated in Figure 2.

The goal of this work is to address these concerns and to investigate whether taking lower bounds into account could improve the quality of the learned model. A first few steps in this direction will be to understand the work done by Bascol et al. (2019), to get familiar with the

[1] It means that is can be expressed as the ratio of two linear functions.

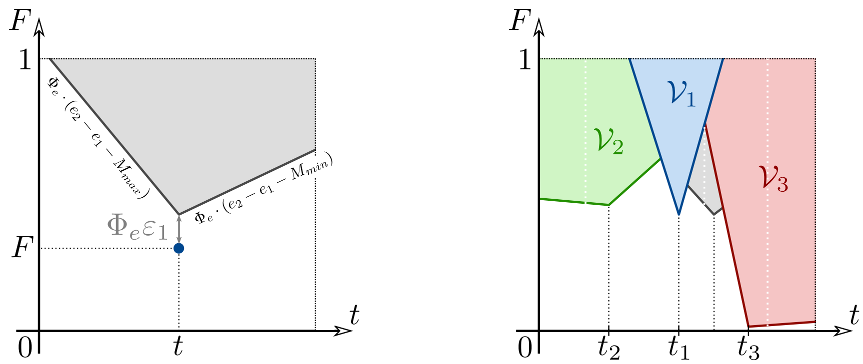

Figure 1: Illustration of the method of Bascol et al. (2019). The theoretical upper bound (left plot) guides the optimization toward models that could maximize the F-measure (right plot).

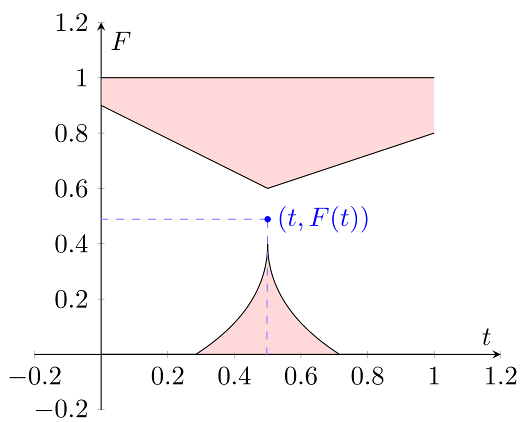

Figure 2: A lower bound may be used to identify models that could perform very poorly.

corresponding code that is available on Github, and to rely on the work of Bascol et al. (2019) and Lohaus et al. (2020) to derive appropriate lower bounds.

Depending on your skills, we can start by extending the work to any pseudo-linear performance measure and see how recent work Mao et al. (2025) can be used to propose a new method that would refine the results of the CONE algorithm.

Keywords: Machine Learning, Imbalanced Data, Classification, Learning Theory

References

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